- *Head-based sampling*—In this case, you decide which span to emit at the beginning of the request (the root span). It's simple to implement, but given that you likely don't know if there will be any issues downstream (nodes and leaves of the tree), you risk missing out on interesting or valuable spans. In addition, given that the sampling decision is made at the start, the decision needs to be globally propagated.
- *Tail-based sampling*—In a sense, it's the opposite of head-based sampling, since the decision is made when the request is completed. That's great, since you know for certain which spans are relevant, but it has the challenge of being, in practice, hard to know when the request is actually completed. For example, one or more services along the request path may time out. How long will you wait until you declare the request completed?

If you're interested in further details on sampling, I highly recommend the book *Cloud-Native Observability With OpenTelemetry* by Alex Boten (Packt, 2022; http://mng.bz/rW9B). With that, let us move on to the topic of resources.

8.3.2 Observability tax

We touched on resource usage a little bit when we talked about dropped spans, at the start of the subsection. Let's step back a bit and look at what I like to call the "observability tax"—that is, costs you will encounter in the context of distributed tracing (though they are not necessarily unique to it—some, if not most, of which you will also see with metrics and logs—but the controls may differ):

- *Instrumentation costs*—Cover everything that has to do with generating spans, including the manual work you have to put into emitting spans from your application, the costs of finding and fixing bugs related to instrumentation, the additional resource usage your app has due to instrumentation (e.g., even in the best case, auto-instrumentation will consume CPU cycles and occupy RAM you have to pay for).
- *Transport and ingestion costs*—Break down into the network ingress, at least in the context of cloud provider accounting (where you typically need to pay for network usage); and sending spans from your environment (that could be on-prem) into the backend, where you may have to deal with costs of double-digit dollars or euros per TB. In addition, your distributed tracing may charge per GB or amount of spans ingested.
- *Storage costs*—Typically, not the highest line item you'd find in your provider bill, since storage is optimized for efficiency and, oftentimes, even offered for free. You usually have limited or fixed retention periods (e.g., AWS X-Ray has 30 days, and Datadog has 15 days).
- *Query costs*—Represent all lookups and API calls to retrieve a range of traces for the purpose of viewing them or presenting them in an aggregate form, including histograms (http://mng.bz/V1AP); service maps (http://mng.bz/x4DW);